

KPI Handbook — Manufacturing Metrics

E-Bike Manufacturing — Enhanced MES Reference

Source dataset: enhanced_mes.db · Document Classification: Internal Use Only · Version 1.0 · July 2026

This handbook defines every performance metric tracked in the MES, the formula behind it, the source tables, and the plant's current baseline computed from the dataset. Use these definitions to keep reporting consistent across shifts and dashboards.

1. Overall Equipment Effectiveness (OEE)

OEE = Availability × Performance × Quality

Source: OEEMetrics (daily roll-up per machine). Window: 2025-04-12 to 2025-07-11.

Plant-wide baseline: Availability 78.4%, Performance 81.0%, Quality 96.1%, giving an **OEE of 62.5%**. Quality is strong; availability is the main lever for improvement, consistent with the downtime profile in Section 3.

- **Availability** = Run Time / Planned Production Time. Lost to breakdowns, changeovers and material waits (Downtimes).
- **Performance** = (Ideal Cycle Time × Count) / Run Time. Reflects speed losses and minor stops.
- **Quality** = Good Count / Total Count. Derived from QualityControl yield.

OEE by machine (world-class reference: 85%)

Machine	Availability	Performance	Quality	OEE	Band
Machine Qua-70	89.6%	86.4%	97.1%	76.6%	Good
Machine Fin-61	85.8%	81.6%	95.9%	68.5%	Good
Machine Qua-71	78.5%	87.5%	97.7%	68.3%	Good
Machine Mot-31	84.6%	79.5%	95.8%	66.0%	Good
Machine Whe-40	79.0%	83.9%	96.3%	65.1%	Good
Machine Bat-20	80.2%	81.6%	97.2%	64.8%	Fair
Machine Fin-60	81.0%	81.2%	96.1%	64.5%	Fair
Machine Pac-80	74.8%	84.9%	96.9%	62.7%	Fair
Machine Bat-21	75.3%	81.8%	97.0%	61.1%	Fair
Machine Mot-30	77.3%	78.6%	95.3%	59.3%	Fair
Machine Fra-11	78.1%	77.0%	95.2%	58.5%	Fair
Machine Pai-50	74.9%	76.4%	94.2%	55.5%	Fair
Machine Fra-12	71.7%	76.9%	95.1%	53.6%	Improve
Machine Fra-10	66.8%	76.9%	95.2%	50.0%	Improve

Frame-welding machines (Fra-10/11/12) sit at the bottom of the OEE table — the same assets carrying vibration anomalies in the condition-monitoring data.

2. Quality: First-Pass Yield, Rework & Scrap

Across **766 inspection checks**, results split as: pass 190, rework 574, fail 2.

- **First-Pass Yield (FPY)** = checks passing first time / total checks = **24.8%**. Target $\geq 90\%$.
- **Average yield rate** (QualityControl.YieldRate) = **96.7%** on passed lots.
- **Scrap** — 2,064 units scrapped against 67,487 planned ($\approx 3.1\%$). Investigate any work center trending $> 3\%$.

Escalation thresholds

- FPY below 90% on any shift → quality review
- A 'fail' result or a severity-5 defect → immediate Root Cause Analysis
- The same DefectType occurring 3× in one shift → stop-and-fix

3. Reliability: Downtime, MTBF & MTTR

Downtime events are split into planned and unplanned. Availability loss is dominated by unplanned stoppages (equipment failure, material shortage, tooling).

Category	Events	Avg Duration	Total Minutes
Planned	25	86 min	2,142
Unplanned	26	89 min	2,309

- **MTBF** (Mean Time Between Failures) = Total Operating Time / Number of unplanned failures. Compute per machine from Downtimes where Category = 'unplanned'.
- **MTTR** (Mean Time To Repair) = Total Repair Time / Number of repairs = mean unplanned Duration (≈ 89 min plant-wide).
- **Downtime %** = Downtime minutes / Planned Production Time, available directly in OEE Metrics.

4. Throughput, Cycle Time & Material Efficiency

- **Cycle time** — derive from WorkOrders (ActualEndTime – ActualStartTime) / ActualProduction; standard per-unit times are held in Products.StandardProcessTime.
- **Throughput** — completed units per period from WorkOrders.ActualProduction; the constraint is Final Assembly (15 bikes/day).
- **Material usage variance** — mean 2.0% (MaterialConsumption.VariancePercent); large negative spikes indicate over-consumption and are worth investigating for scrap or theft.
- **Schedule adherence** — compare Planned vs Actual start/end times on WorkOrders.

All KPI baselines in this handbook are computed live from enhanced_mes.db and should be recomputed whenever the dataset is refreshed.